🧪 Step-by-Step Procedure

1 Login to AWS

- Go to: https://aws.amazon.com
- Sign in to **AWS Management Console**

2 Go to EC2 Dashboard

- Search: **EC2**
- Open EC2 service
- Click "**Launch Instance**"

3 Choose AMI (Operating System)

- Select:
  - **Amazon Linux (recommended)** OR
  - Ubuntu

👉 AMI = **Amazon Machine Image (OS template)**

4 Choose Instance Type

- Select:
  - `t2.micro` (Free tier)

👉 Viva point:

> Instance type defines CPU, RAM, and performance.

5 Create / Select Key Pair 🔑

- Click **Create new key pair**
- Choose:
  - Type: RSA
  - Format: `.pem`
- Download it (VERY IMPORTANT)

👉 Used for **secure SSH login**

6️⃣ Configure Network Settings

- Allow:
  - ✅ SSH (port 22)
  - ✅ HTTP (port 80)

👉 Security group = firewall rules

7️⃣ Launch Instance 🚀

- Click **Launch Instance**

8️⃣ Connect to Instance

- Select instance → Click **Connect**
- Use SSH command:

```
ssh -i "your-key.pem" ec2-user@your-public-ip
```

9️⃣ Verify Instance

Run:

```
ls
pwd
```

🎯 Expected Output (What Examiner Looks For)

- Instance is in **running state**
- You can:
  - Connect via SSH
  - Execute commands

🧠 Important Viva Questions

❓ What is EC2?

> Virtual server in cloud.

❓ What is AMI?

> Template used to create VM (OS + software).

❓ What is Key Pair?

> Used for secure login (public + private key).

❓ What is Security Group?

> Acts as a firewall controlling inbound/outbound traffic.

❓ What is t2.micro?

> Free tier instance with limited CPU & RAM.

❓ Difference: EC2 vs Traditional Server

EC2	Traditional
On-demand	Fixed hardware
Scalable	Hard to scale
Pay-as-you-go	High upfront cost

⚠️ Common Mistakes (Avoid Tomorrow)

- ❌ Forgetting to download key pair
- ❌ Not enabling SSH in security group
- ❌ Wrong username (`ec2-user` vs `ubuntu`)
- ❌ Instance not in running state

💡 Extra (if they ask to install something)

After connecting:

```
sudo yum update -y
sudo yum install httpd -y
sudo systemctl start httpd
```

👉 Then open browser:

```
http://your-public-ip
```